Final Project Report

Pipelined IEEE-754 Double-Precision Floating-Point Multiplier

Operating Frequency: 2.0 GHz　(Clock Period = 0.5 ns)

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1. Architecture & Algorithm

1.1 IEEE 754 雙精度浮點數格式回顧

數值格式為 IEEE 754 double precision，每筆數值固定為 64 bits，欄位定義依下表：

欄位	寬度	意義
Sign (s)	1 bit	0 為正、1 為負
Exponent (e)	11 bits	bias = 1023
Fraction (f)	52 bits	尾數的小數部分

normal number 的數值為： **value = $(-1)^s \times 1.f \times 2^{(e-1023)}$**

依 e、f 之組合分類為以下五種 case，硬體設計時須個別處理：

類別	biased exp (e)	fraction (f)	數值意義
Zero	0	0	±0
Subnormal	0	≠ 0	$(-1)^s \times 0.f \times 2^{-1022}$
Normal	1 ~ 2046	任意	(−1)ˢ × 1.f × 2^(e−1023)
Infinity	2047	0	±∞
NaN	2047	≠ 0	Not a Number

對 Subnormal 採 Flush-To-Zero (FTZ) 處理

1.2 浮點數乘法演算法

給定兩個數值 A、B，乘法結果 Z = A × B 由以下三組運算構成：

- **Sign 運算**：s = s_a XOR s_b
- **Exponent 運算**：e = e_a + e_b − 1023 (扣掉多一次的 bias)
- **Mantissa 運算**：m = (1.f_a) × (1.f_b) (53-bit × 53-bit)

Round-to-Nearest-Even 規則

下方表列 round_up 條件，其中 G 為 guard bit、 S 為 sticky bit：

條件	捨入方向	說明
G = 0	Round down	小於 1/2 ，截斷
G = 1 且 (S=1 或 LSB=1)	Round up	>= 1/2 但需 round to even
G = 1 且 S=0 且 LSB=0	Round down	Round-to-even 之 tie 情況

1.3 53×53 尾數乘法器拆解 (27 × 26 Split)

將 53-bit 尾數從中拆成「上 27 bits + 下 26 bits」兩段：

m_a = {a_h[26:0], a_l[25:0]}, m_b = {b_h[26:0], b_l[25:0]}

乘積依據分配律展開為四項：

Partial product	size	位移權重
pp_hh = a_h × b_h	27 × 27 → 54 bits	× 2^{52}
pp_hl = a_h × b_l	27 × 26 → 53 bits	× 2^{26}
pp_lh = a_l × b_h	26 × 27 → 53 bits	× 2^{26}
pp_ll = a_l × b_l	26 × 26 → 52 bits	× 2^{0}

1.4 FSM 與 8 pipeline stages

FSM (state[1:0]，0=LOAD、1=COMP、2=OUT)：

Stage	行為
C1	sign / exp / fraction 拆出；implicit 1 補回；ZERO / INF / NaN 偵測
C2	Sign XOR；e_sum = e_a + e_b − 1023；mantissa 切成上 27 bits + 下 26 bits
C3	Partial product：pp_hh (27×27)、pp_hl (27×26)、pp_lh (26×27)、pp_ll (26×26)
C4	pp_mid = pp_hl + pp_lh
C5	upper_combo = pp_hh + pp_mid[53:26]；lower_combo =(pp_mid[25:0] + pp_ll
C6	合併成 prod_full；正規化 (leading 1)；產出 mant、guard、sticky、e_norm
C7	Round-to-Nearest-Even：round_up = G & (S | LSB)；mantissa overflow 則右移並 e_final += 1
C8 (組合)	Special-case 優先 ； 64-bit pack

2. Test

2.1 self checking

real_Z = real_A * real_B;

C = $realtobits(real_Z);

可得到任意輸入對應的乘法結果

```
real_A = $bitstoreal(A);
real_B = $bitstoreal(B);
real_Z = real_A * real_B;
C = $realtobits(real_Z);
```

```
while(READY !== 1'b1)
    @(negedge CLK);

for(i=0; i<8; i=i+1)
begin
    Z[(i*8) +: 8] = DATA_OUT;
    if(i<7)
        @(negedge CLK);
end
```

```
if(Z !== C)
begin
    err_count = err_count + 1;
    $display("[FAIL]");
    $display("Expected : %b", C);
    $display("Actual   : %b\n", Z);
end
else
begin
    $display("[PASS]");
    $display("Expected : %b", C);
    $display("Actual   : %b\n", Z);
end
```

2.2 test pattern

透過 {$random, $random} 產生 64-bit 浮點數作為 test pattern，每筆測資皆透過 if (Z !== C) err_count++ 自動比對。模擬結束時 testbench 依據 err_count 顯示 simulation fail/pass

```
// set test pattern
$display("A = 1.5,  B = 2.0");
auto_test($realtobits(1.5), $realtobits(2.0));
$display("A = -3.14,  B = 2.5");
auto_test($realtobits(-3.14), $realtobits(2.5));
$display("A = 0.3333333333333333,  B = 3.0");
auto_test($realtobits(0.3333333333333333), $realtobits(3.0));



// random test patten
$display("------random test pattern------");
repeat(20)
begin
    rand_A = {$random, $random};
    rand_B = {$random, $random};
    $display("A = %.17e,  B = %.17e", $bitstoreal(rand_A), $bitstoreal(rand_B));
    auto_test(rand_A, rand_B);
end

// special case test pattern
$display("A = 0.0,  B = 3.0");   // ZERO
auto_test(64'h0000000000000000, $realtobits(3.0));
$display("A = INF,  B = 2.0");   // INF
auto_test(64'h7FF0000000000000, $realtobits(2.0));
$display("A = NaN,  B = 2.0");   // NaN
auto_test(64'h7FF8000000000000, $realtobits(2.0));
$display("A = INF,  B = ZERO");  // INF * ZERO = NaN
auto_test(64'h7FF0000000000000, 64'h0000000000000000);
$display("A = Max,  B = 2.0");   // overflow -> INF
auto_test(64'h7FEFFFFFFFFFFFFF, $realtobits(2.0));
$display("A = Min,  B = Min");   // underflow -> ZERO
auto_test(64'h0010000000000000, 64'h0010000000000000);


// report
$display("==========================================================");
if(err_count != 0)
    $display("Simulation Fail!");
else
    $display("Simulation OK!!!!");
$display("==========================================================");
```

2.3 Special Case Test

針對下列 special case 設計 test pattern，與預期結果比對：

測試項目	輸入 A × B	預期輸出 (IEEE-754)	結果
Zero	0.0 × 3.0	0	PASS
Infinity	Inf × 2.0	Inf	PASS
NaN	NaN × 2.0	NaN	PASS
Inf × Zero	+Inf × 0.0	NaN	PASS
Overflow	DBL_MAX × 2.0	Inf	PASS
Underflow	min_normal × min_normal	0 (Flush-To-Zero)	PASS

3. Synthesis

- **timing report**

  Timing slack = 4ps

```
Cost Group   : 'CLK' (path_group 'CLK')
Timing slack :       4ps
Start-point  : b_l_c2_reg[1]/CP
End-point    : pp_ll_c3_reg[38]/D
```

- **area report**

  total area = 8145.088

```
====================================================================
  Generated by:           Genus(TM) Synthesis Solution 20.10-p001_1
  Generated on:           Jun 03 2026  04:11:41 pm
  Module:                 CHIP
  Technology library:     N16ADFP_StdCellss0p72v125c_ccs 100
  Operating conditions:   ss0p72v125c
  Interconnect mode:      global
  Area mode:              physical library
====================================================================

Instance Module  Cell Count  Cell Area  Net Area   Total Area
--------------------------------------------------------------------
CHIP                 11142   5844.234  2300.854     8145.088
```

4. APR

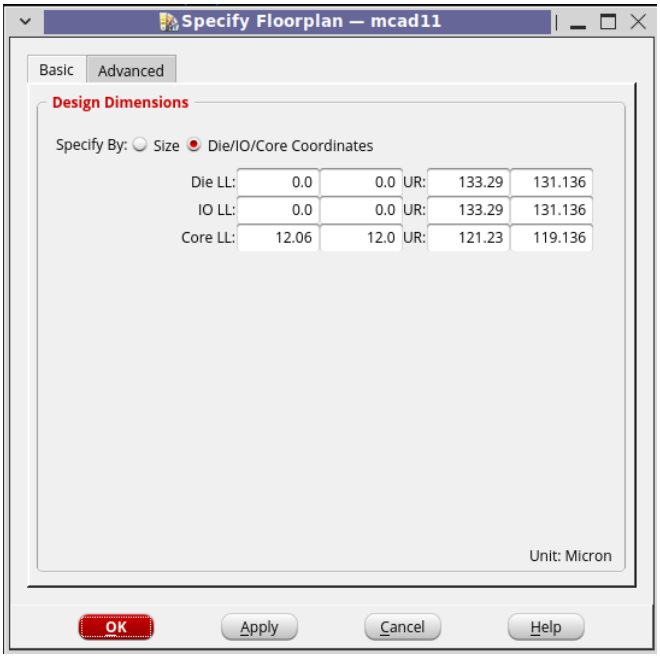
- **chip size**

  Die = 133.29 x 131.136

  Core width = 121.23 - 12.06 = 109.17

  Core height = 119.136 - 12.0 = 107.136

  Core area = core width x core height =109.17 × 107.136 ≈ 11,694.4



- **postRoute**
  - setup time WNS = 0.064ns

```
Path 1: MET Setup Check with Pin pp_lh_c3_reg[47]/CP
Endpoint:   pp_lh_c3_reg[47]/D (^) checked with  leading edge of 'CLK'
Beginpoint: a_l_c2_reg[13]/Q   (v) triggered by  leading edge of 'CLK'
Path Groups: {reg2reg}
Analysis View: CHECK_SETUP_TIME
Other End Arrival Time          0.065
- Setup                         0.013
+ Phase Shift                   0.500
+ CPPR Adjustment               0.000
= Required Time                 0.552
- Arrival Time                  0.487
= Slack Time                    0.064
     Clock Rise Edge                     0.000
     + Source Insertion Delay           -0.067
     = Beginpoint Arrival Time          -0.067
```

  - hold time WNS = 0.017ns

```
Path 1: MET Hold Check with Pin b_h_c2_reg[21]/CP
Endpoint:   b_h_c2_reg[21]/D (v) checked with  leading edge of 'CLK'
Beginpoint: m_b_c1_reg[47]/Q (v) triggered by  leading edge of 'CLK'
Path Groups: {reg2reg}
Analysis View: CHECK_HOLD_TIME
Other End Arrival Time          0.060
+ Hold                          0.008
+ Phase Shift                   0.000
- CPPR Adjustment               0.055
= Required Time                 0.013
  Arrival Time                  0.029
  Slack Time                    0.017
     Clock Rise Edge                     0.000
     + Source Insertion Delay           -0.055
     = Beginpoint Arrival Time          -0.055
```

- **total power**

  Internal power = 11.08

  Switching power = 4.941

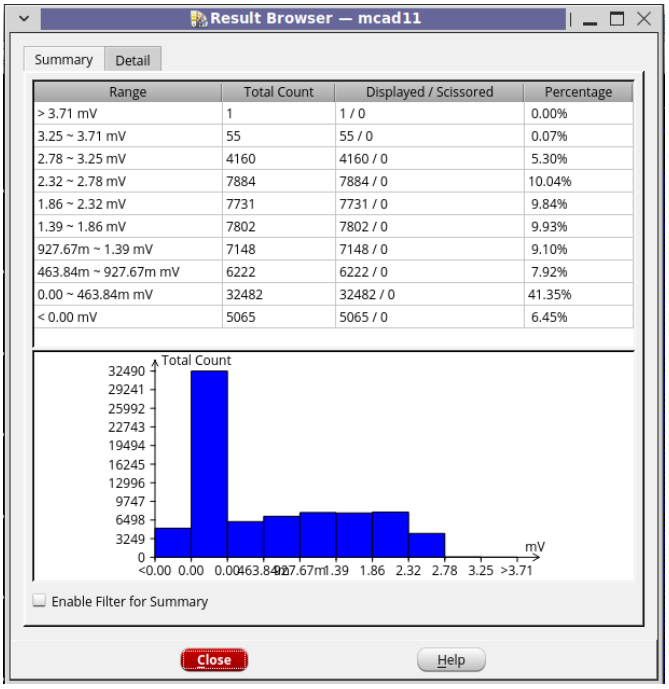
  Total power = 18.09

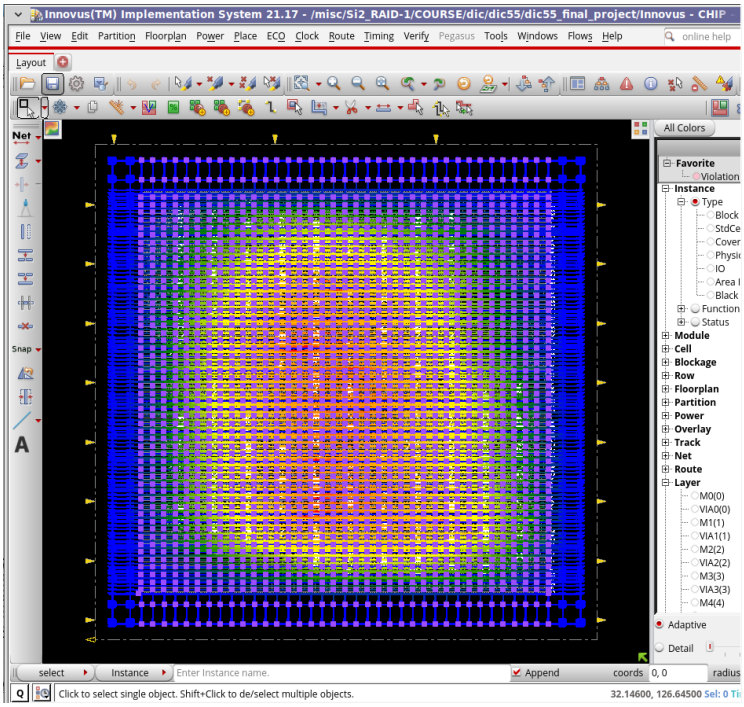
  Leakage power = 2.068

```
Total (  19602 of  19602  )       11.08       4.941      18.09       2.068
Total Capacitance              4.376e-11 F
```

- **IR drop analysis**

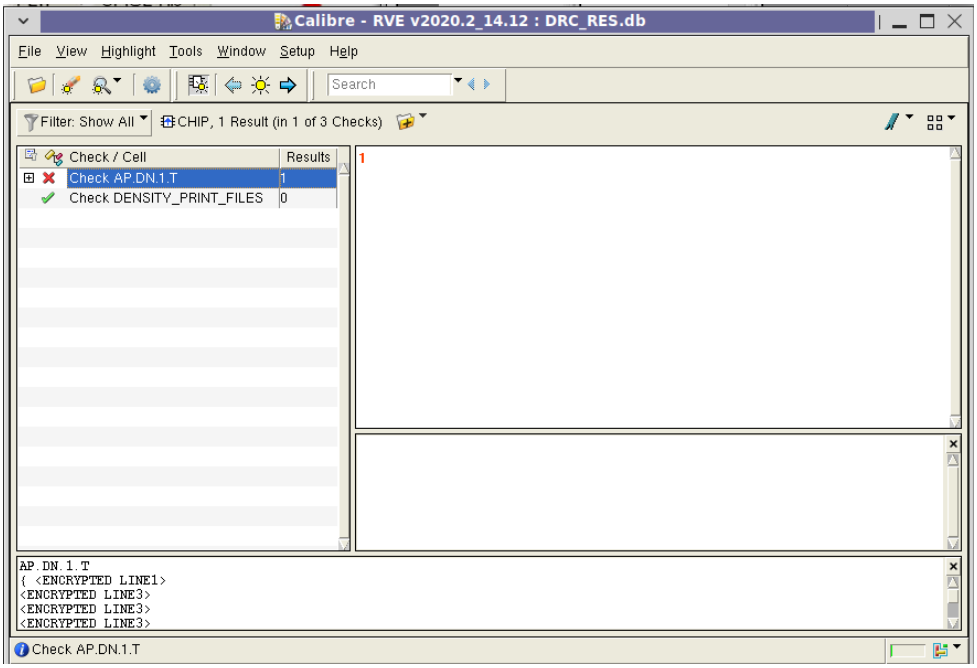
  data range: <0.00 mV to >3.71mV



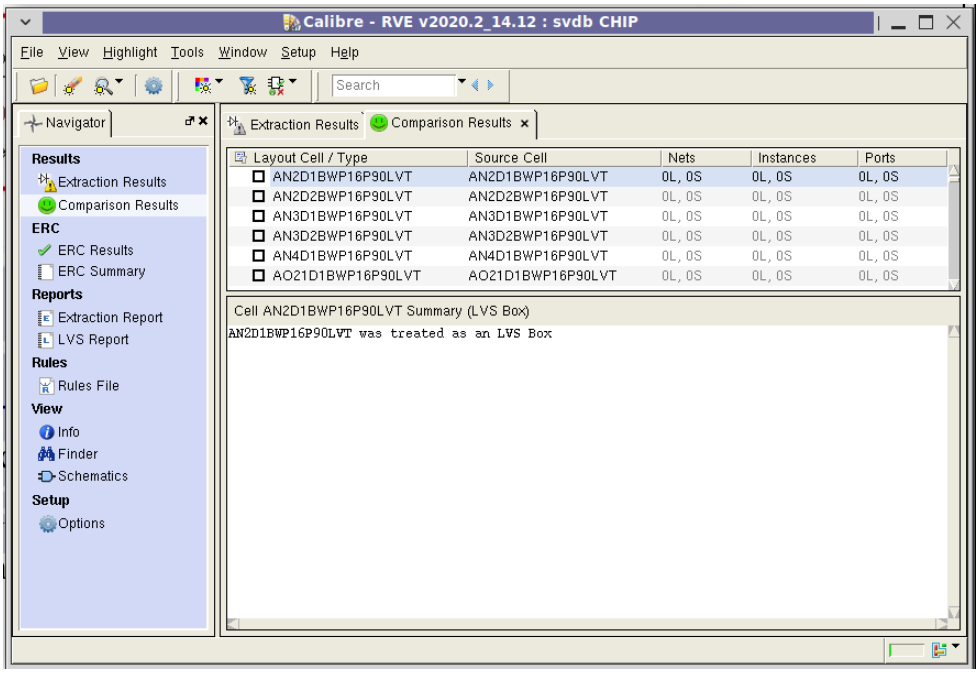


5. DRC & LVS

- **DRC PASSED**



- **LVS PASSED**


Calibre - RVE v2020.2_14.12 : svdb CHIP
File View Highlight Tools Window Setup Help
Search
Navigator
Results
Extraction Results
Comparison Results
ERC
ERC Results
ERC Summary
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Extraction Results
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
AN2D1BWP16P90LVT AN2D1BWP16P90LVT 0L, 0S 0L, 0S 0L, 0S
AN2D2BWP16P90LVT AN2D2BWP16P90LVT 0L, 0S 0L, 0S 0L, 0S
AN3D1BWP16P90LVT AN3D1BWP16P90LVT 0L, 0S 0L, 0S 0L, 0S
AN3D2BWP16P90LVT AN3D2BWP16P90LVT 0L, 0S 0L, 0S 0L, 0S
AN4D1BWP16P90LVT AN4D1BWP16P90LVT 0L, 0S 0L, 0S 0L, 0S
AO21D1BWP16P90LVT AO21D1BWP16P90LVT 0L, 0S 0L, 0S 0L, 0S
Cell AN2D1BWP16P90LVT Summary (LVS Box)
AN2D1BWP16P90LVT was treated as an LVS Box